



Teaching Business Process Simulation and Sustainability in Procurement

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BPM & Business Process Simulation

Business Process Management (BPM): Management discipline based on the cross-organizational analysis, design, and implementation of processes (Rosemann & vom Brocke 2015)

Business Process Simulation (López-Pintado & Dumas 2022, Chapela-Campa et al. 2023):

Technique to

- analyze “what-if” scenarios based on process models that include parameters capturing available resources, e.g., in terms of capacity, processing times, or costs
- address questions about business process cycle and waiting times, gaining valuable insights into potential improvements and bottlenecks to refine strategies and achieve optimal performance

Free registration for SAP Signavio Academic Workspace (includes SAP Signavio Process Manager, including BPMN Process Simulation): <https://academic.signavio.com/p/register>

* López-Pintado, O. & Dumas, M. (2022): Business Process Simulation with Differentiated Resources: Does it Make a Difference? In: *Business Process Management*. Cham: Springer, 361–378.

Rosemann, M. & vom Brocke, J. (2015): *Handbook on Business Process Management I: Introduction, Methods, and Information Systems*. Berlin, Heidelberg: Springer.

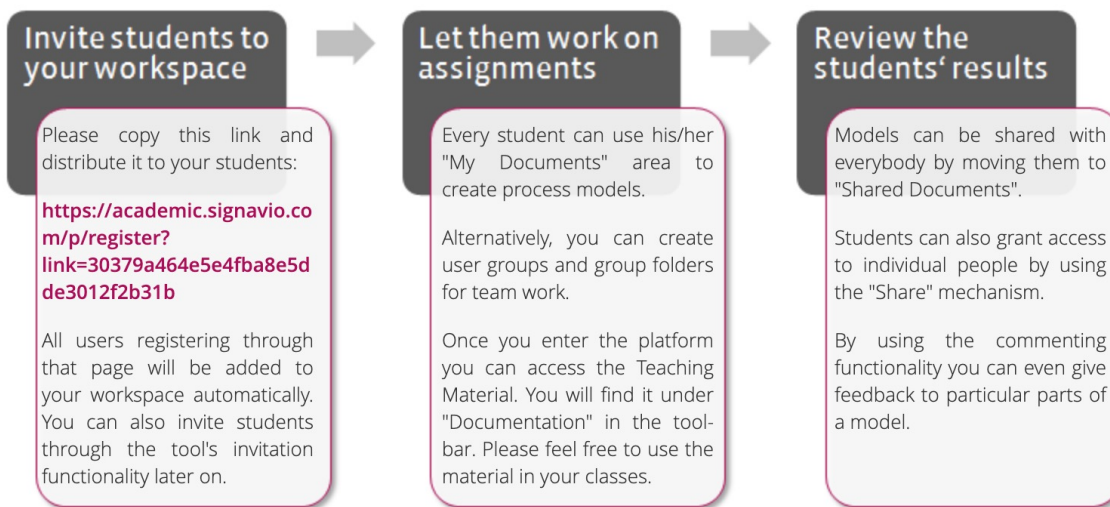
Chapela-Campa, D., Benckroun, I., Baron, O., Dumas, M., Krass, D. & Senderovich, A. (2023): Can I Trust My Simulation Model? Measuring the Quality of Business Process Simulation Models.

SAP Signavio Process Transformation Suite, Academic Edition

Live demo & intro to SAP Signavio

Free workspace available at <https://academic.signavio.com/p/register>

More information can be found at <https://www.signavio.com/academic-and-research-alliances/>



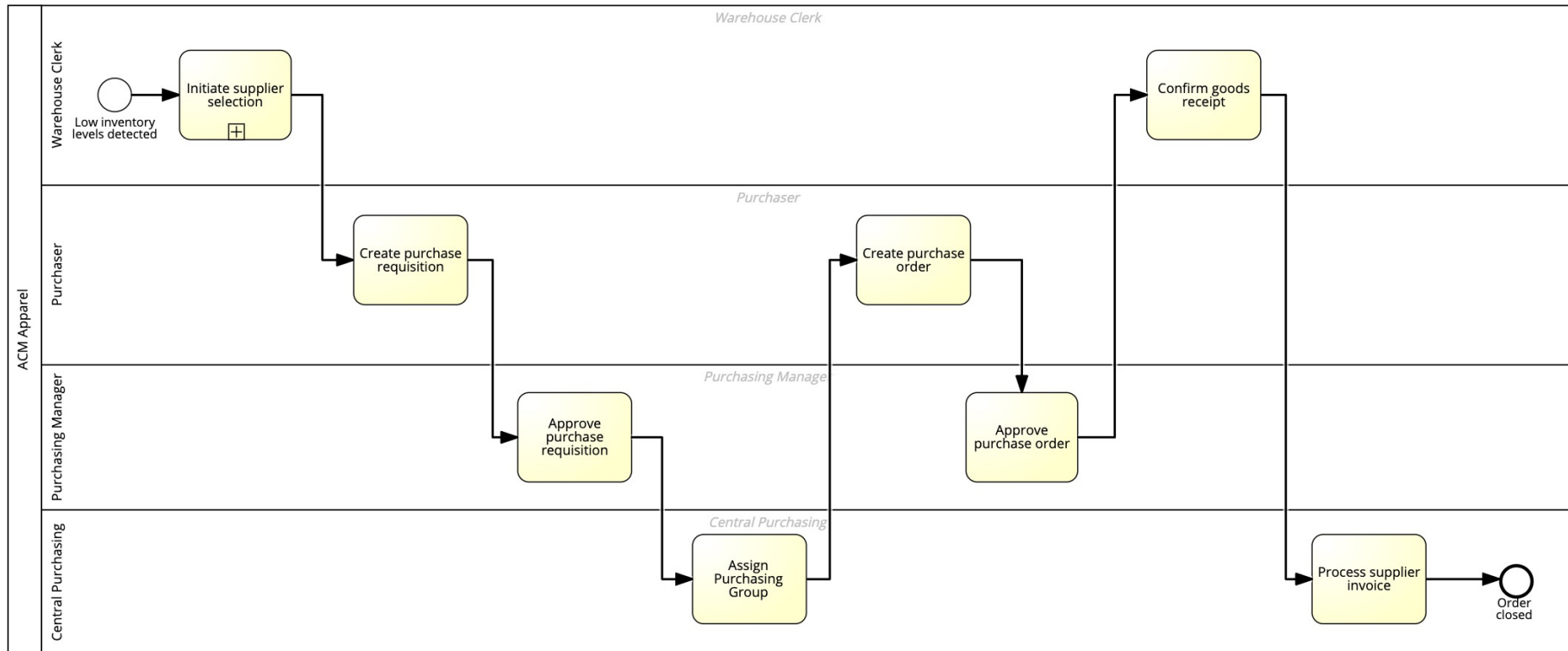
Case Study: ACM Apparel

A Company that **Makes Apparel** (ACM Apparel) designs and prints t-shirts, dresses, pants and other clothing items for resale, mostly through European locations. It is headquartered in London, UK, and employs over 1.000 employees.

1) ACM Apparel's **Source-to-Pay (S2P) Process** starts with the Warehouse Clerk initiating supplier selection, followed by the creation of a purchase requisition by the Purchaser. After the Purchasing Manager has approved the purchase requisition, the Central Purchasing department assigns the applicable purchasing group. Once the purchasing group is assigned, the Purchaser creates the Purchase Order (PO). The purchase order has to be approved by the Purchasing Manager. Once the purchased goods have arrived, the Warehouse Clerk confirms the goods receipt, after which Central Purchasing proceeds to process the supplier invoice. This step concludes the S2P process.

Use the **Graphical Editor** to model the process in **BPMN 2.0!**

1) ACM Apparel's S2P Process Model

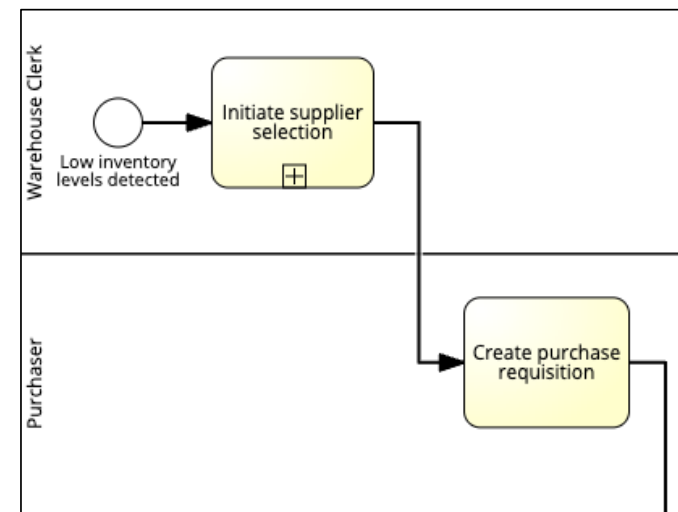
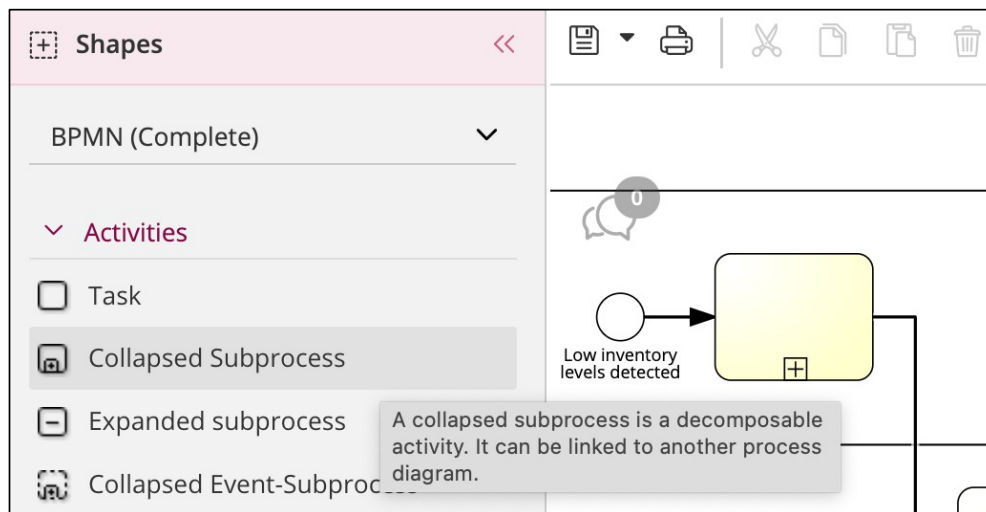


→ Simulation case study for S/4HANA S2P automation available but not focus of this presentation.

2) ACM Apparel's S2P Process Model

2) ACM Apparel's **supplier selection** follows a subprocess of the S2P process.

→ In the activity types, select “Collapsed Subprocess”



Case Study: ACM Apparel **Sustainability in S2P**

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2) ACM Apparel's **supplier selection** can be modeled as a **subprocess** of the overall S2P process. After the Warehouse Clerk assessed inventory needs, the Procurement department specifies the necessary terms and conditions for purchase of new inventory. Afterward, the Procurement department has to find potential suppliers (e.g., through web-based search and internal contacts) and evaluate them. After creating a so-called request for proposal (RFP), the Procurement department solicits offers by a number of suppliers. Procurement collects and evaluates the offers to then make a final supplier and order selection. The final selection of a supplier and respective order concludes the process.

Use the **QuickModel functionality** (pop-up module selection menu upon click on name icon) to create the supplier selection process in **BPMN 2.0**.

Case Study: ACM Apparel **Sustainability in S2P**

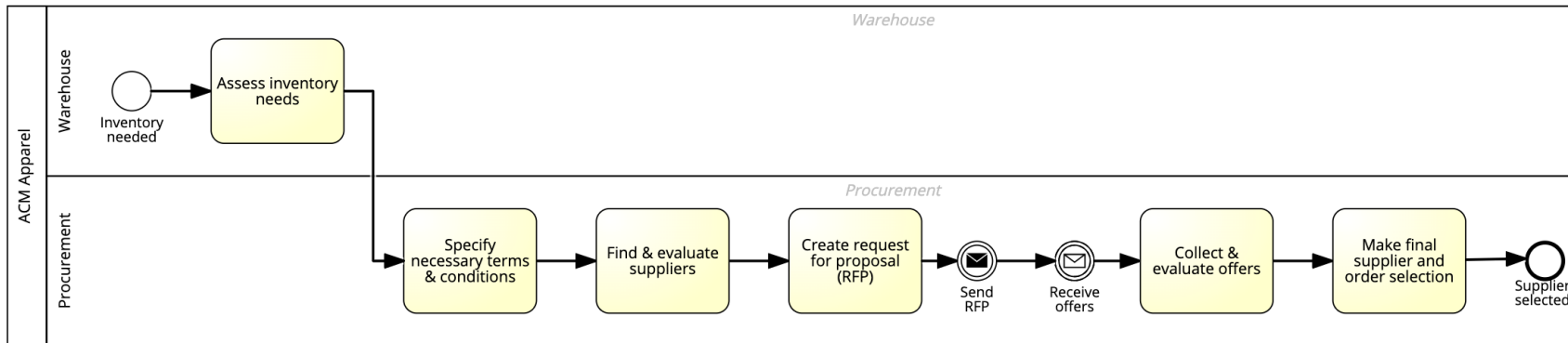
A Company that **M**akes **A**pparel (ACM Apparel) designs and prints t-shirts, dresses, pants and other clothing items for resale, mostly through European locations. It is headquartered in London, UK, and employs over 1.000 employees.

3) In the QuickModel functionality, add the following execution times to the process activities:

⊕ Add activity above 🗑 Remove activity 🔗			
	What?	Who?	Execution Time
1	Assess inventory needs	Warehouse Clerk	10
2	Specify necessary terms & conditions	Purchaser	30
3	Find & evaluate suppliers	Purchaser	10
4	Create request for proposal (RFP)	Purchaser	30
5	Evaluate & compare offers	Purchaser	10
6	Select supplier	Purchaser	5

Since the Procurement team is manually selecting suppliers and solicits offers via e-mail (RFP), ACM Apparel must wait for supplier proposals before proceeding with offer selection. **Navigate to the Graphical Editor (pop-up module selection menu upon click on name icon) to add these components to the sourcing process!**

Case Study: ACM Apparel **Sustainability in S2P**



4) From the Graphical Editor, navigate to the Simulation module (pop-up module selection menu upon click on name icon). Scroll down to Scenarios. **Since manual sourcing entails printing and postage costs, add an extra 10€ to the task **offer collection & evaluation**.**

In this manual sourcing process, the Procurement team can only proceed with offer collection & evaluation once suppliers have responded to the RFP. **To account for this, add an average of 4:00 hours waiting time to the **Intermediate Message Event** in the **Duration** tab.**

Make sure that both Warehouse Clerk and Procurement are scheduled on 40-hour work weeks with 50€ associated costs per week.

SAP Ariba & SAP Business Network: Sustainability in S2P

Background: [EU's Sustainability Due Diligence Directive \(SDDD\)](#) (Supply Chain Act) “applies to the company's own operations, their subsidiaries and their value chains (direct and indirect established business relationships). In order to comply with the corporate due diligence duty, companies need to: [...] identify actual or potential adverse human rights and environmental impacts [...].”

		LARGE EU LIMITED LIABILITY COMPANIES	NON-EU COMPANIES	SMALL AND MEDIUM ENTERPRISES
GROUP 1	500+ employees and more than €150 million of turnover*	+/- 9,400 companies	+/- 2,600 companies	Micro companies and small and medium enterprises (SMEs) are not directly concerned by the proposed rules. However, the proposal provides supporting measures for SMEs, which could be indirectly affected.
GROUP 2	250+ employees and more than €40 million of turnover*, operating in defined high impact sectors such as textiles, agriculture, extraction of minerals. The rules will apply to this group 2 years later than to group 1.	+/- 3,400 companies	+/- 1,400 companies	

*Worldwide turnover for EU companies, and EU-wide turnover for non-EU companies

Important elements in ACM Apparel's Corporate Sustainability Due Diligence:

- Labour rights
 - No land grabbing & right to form unions
 - ...
- Climate & environmental protections
 - Transparency & reporting
 - Adherence to Paris Agreement
 - ...

Case Study: ACM Apparel **Sustainability in S2P**

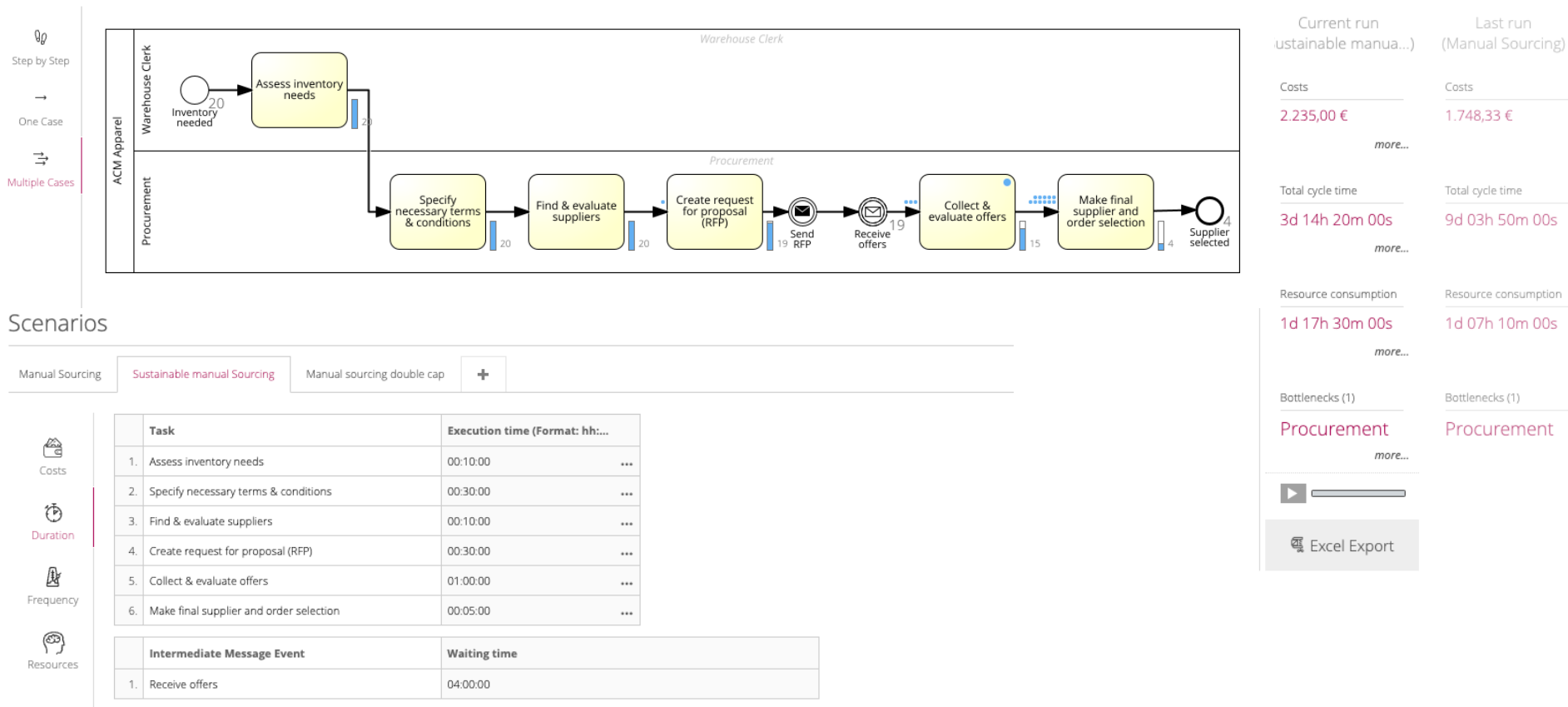
A Company that **M**akes **A**pparel (ACM Apparel) designs and prints t-shirts, dresses, pants and other clothing items for resale, mostly through European locations. It is headquartered in London, UK, and employs over 1.000 employees.

5) Compliance with the Supply Chain Act of the European Union requires ACM Apparel to ensure sustainable practices of suppliers in their value chain. With the manual sourcing process in place, this increases the time needed to evaluate and compare offers from previously 10 minutes to one hour.

Adjust the duration settings in **Simulation** mode and re-run the simulation to compare scenarios.

What could be done to resolve the bottleneck?

Compliance with Supply Chain Act requires additional resources



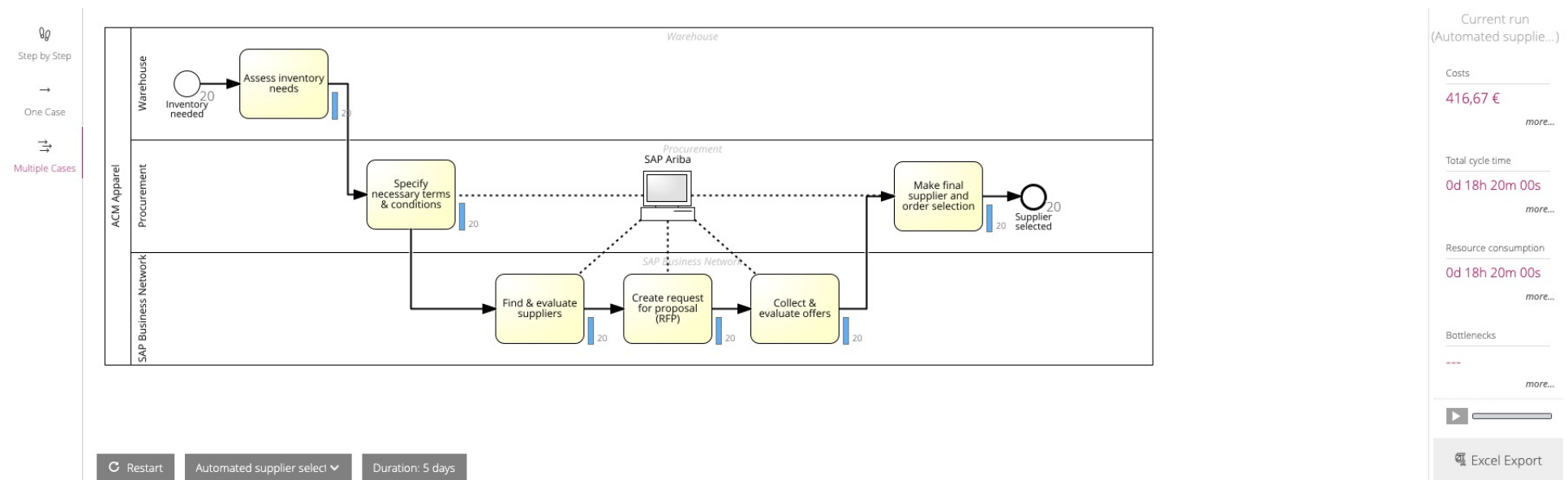
Automation and Sustainability Optimization

SAP Ariba, a cloud-based procurement solution for interactions between buyers & sellers/suppliers, in combination with **SAP Business Network**:

- Removes complexity in procurement processes
- Creates supply chain transparency & sourcing compliance:
 - Supplier self-reported data
 - External certifications
 - Sustainability metrics
- Standardizes & automates procurement processes (e.g., ERP data integration)

→ Re-model the process to automate supplier selection with **SAP Ariba & SAP Business Network** to resolve the bottleneck, decrease cost, and ensure ESG compliance!

Graphical Editor: Introduce SAP Ariba & SAP Business Network



Scenarios

Automated supplier selection +

	Task	Execution time (Format: hh:mm:ss)	
1.	Assess inventory needs	00:10:00	...
2.	Specify necessary terms & conditions	00:10:00	...
3.	Find & evaluate suppliers	00:10:00	...
4.	Create request for proposal (RFP)	00:10:00	...
5.	Collect & evaluate offers	00:10:00	...
6.	Make final supplier and order selection	00:05:00	...

Remove scenario Save scenario

Summary: Teaching Business Process Simulation and **Sustainability in S2P** with SAP Signavio, SAP Ariba & SAP Business Network

Through automation of supplier selection process, ACM Apparel manages increased due diligence requirements for procurement, resolves bottlenecks, and ensures compliance with the SCA

→ Supplier selection process costs reduced to less than one third of manual supplier selection process.



[SAP Business Network](#) creates supply chain transparency & sourcing compliance.



[SAP Ariba](#) standardizes & automates procurement processes (e.g., ERP data integration) and ensures compliance with sourcing guidelines.

Skills & competencies:

- BPM / BPMN with SAP Signavio
- Business Process Simulation
- Sourcing & procurement knowledge
- Sustainability & SCA compliance
- SAP Ariba & SAP Business Network

Thank you!

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